

Demo: Brain-Controlled Robot for Assisting Basic Upper Limb Tasks

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Abstract—The paper aims to address the challenges faced by individuals with neurological diseases, such as Parkinson’s or multiple sclerosis, who may lose control of their upper limbs. This limitation hampers their ability to perform basic tasks like grasping or releasing objects. Using the MUSE 2[®] EEG sensing headband device, we developed a proof of concept for a brain-controlled robot. By capturing EEG signals originating from the brain and analyzing their frequency domain and compositions, we successfully classified the signals and enabled real-time tasks to be executed on a Raspberry Pi-based robotic arm. Our approach utilized blinking and jaw clenching to reproduce actual gestures on the robot, offering assistance to individuals with compromised upper limb control. The system was evaluated using a Likert survey administered to nine individuals, providing valuable insights into the usability and effectiveness of the brain-controlled robot concept

Index Terms—BCI, Robotics, EEG.

I. INTRODUCTION

Upper limb disabilities, resulting from neurological conditions such as Parkinson’s disease, multiple sclerosis, or other related disorders, pose significant challenges to individuals in performing essential everyday tasks. According to recent statistics, it is estimated that more than 2 million of the population in the United States experience neurological disabilities. In the pursuit of innovative solutions, researchers have explored the utilization of electroencephalography (EEG) technology to restore upper limb functionality. Previous studies have demonstrated the potential of EEG-based systems to enable individuals with upper limb disabilities to control robotic devices using their brain signals. In this context, our project offers a unique contribution by developing a cost-effective, portable, and user-friendly solution, by employing the Muse 2 EEG sensor.

A. System Design

Fig. 1 depicts the comprehensive functional diagram of our project, illustrating the key stages involved.

B. Analytical Component

The Muse 2 sensor consists of 4 electrodes, each delivering 5 frequency components (alpha, beta, gamma, delta, theta). To distinguish between different motions or the absence of motion, we employed a high threshold for blinking and jaw

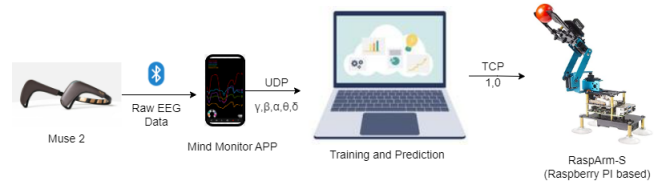


Fig. 1. Functional diagram of implementation

clenching. This provides a reliable basis for action detection within our system.

C. Experimental Evaluation

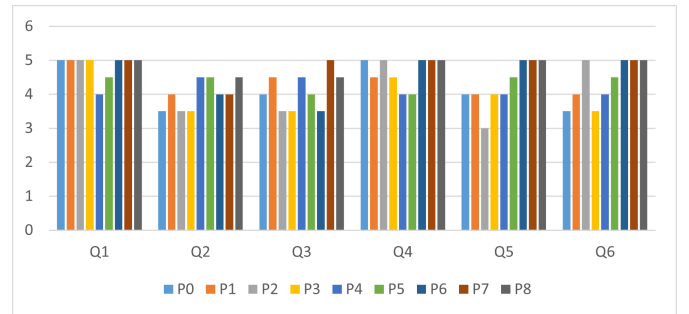


Fig. 2. Likert Experimental Results

We conducted a Likert scale survey to evaluate the user experience and perceptions of our system (Fig. 2). The cohort comprised nine participants (P0 to P8), consisting of five females (F) and four males (M). Their ages ranged from 25 to 31 years. Six survey questions were considered (Q1-Q6).

Overall, the survey results demonstrate a positive response from participants regarding the system’s usability, accuracy, responsiveness, comfort, and potential impact on individuals with upper limb disabilities. This valuable feedback will inform future enhancements and optimizations to further improve the system’s performance and user experience.

II. CONCLUSION

In this project, we have addressed the problem of limited upper limb functionality, offering a solution through our brain-controlled robot system.